

NEERAJ V PATTANASHETTI

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Professional Summary

Graduate Computer Science student specializing in intelligent systems, scalable backend, and full-stack development. Skilled in Python, Java, and JavaScript with hands-on experience in building LLM-based applications, workflow automation, and data-driven AI solutions. Strong foundation in algorithms, machine learning, and distributed architectures, with a passion for developing practical, reliable systems that blend AI reasoning with real-world impact.

Education

Master of Science University of Illinois Chicago	Aug, 2025 – May, 2027
Major in Computer Science Expected Graduation: May, 2027	Chicago, USA
Relevant Coursework: Data Structures & Algorithms, Database Systems, Web Development	
Bachelor of Engineering Visveswaraya Technological University	Dec, 2021 – Jul, 2025
Major in Computer Science GPA: 3.52	Bengaluru, India
Relevant Coursework: Distributed Systems, Machine Learning & Deep Learning, Computer Networks	

Skills

Programming Languages: TypeScript | JavaScript | Python | Java | C++ | C# | Swift (basic)
Frameworks: Node.js | Express | React.js | TensorFlow | PyTorch
Systems & Cloud: AWS | Firebase | CI/CD | Docker | Microservices | Distributed Systems
AI / Machine Learning: Large Language Models (LLMs) | Conversational AI | LangChain | API | Reinforcement Learning | Ray | Deep Learning
Databases: SQL | NoSQL (Firebase Realtime DB) | Data Modelling
Communication: Technical Documentation | Research Writing | Cross-functional Collaboration
Tools/Other: Agile | Jira | Debugging | Version Control | Testing Automation | Docker | Unity 2D/3D

Work Experience

Software Engineering Intern KreedaLoka	Jul, 2023 – Dec, 2023
Bengaluru, India	
ChessEra: A chess app used by Chess academies to improve their student’s skill and also hosts Chess Tournaments.	
- Designed and deployed Arena Mode in ChessEra (Java backend + Unity C# client), improving tournament engagement by 25% and supporting 100+ concurrent users in real time. - Refactored distributed match-management system using object-oriented design principles, improving latency by 40% and enhancing fault tolerance with asynchronous messaging and caching. - Optimized SQL schema and queries, cutting data retrieval 40% and ensuring real-time leaderboard accuracy. - Contributed to Agile sprints code reviews, documentation, with team-based feature development, and system performance monitoring for scalable cloud releases.	

Projects

AI Conversational Avatar	Aug 2025
- Developed an AI-powered 3D avatar integrating LLMs (Gemini) for voice-enabled intelligent interactions, backed by RAG pipelines with Pinecone vector database for semantic context retrieval. - Integrated speech-to-text and text-to-speech APIs for real-time interaction, showcasing conversational AI and web integration. - Prototyped using LLMs, RAG, and Pinecone, demonstrating strong alignment with enterprise AI retrieval and knowledge systems.	
FixMyIot - IoT Troubleshooter	Jan 2025
- Built an AI-powered troubleshooting assistant (React.js, Node.js, Express) using the DeepSeek R1 model and RESTful APIs; reduced manual debugging effort 60%. - Implemented secure authentication, cloud-ready architecture, and CI/CD pipelines with fault-tolerant scaling. - Focused on building a scalable application architecture to handle real-time troubleshooting requests efficiently.	
Hospital Connect	Oct 2024
- Built a React.js healthcare platform integrating Google Maps API + Firebase, enabling users to locate nearby hospitals, check real-time resource availability, and access secure virtual consultations. - Designed real-time chat and authentication system, scaling access for 1,000+ users with robust, fault-tolerant performance. - Built fault-tolerant backend and optimized data handling using distributed principles.	
Facial And Gesture Recognition for Sign Language	Jul 2024
- Implemented a deep learning system with pipeline combining EfficientNetB0 + YOLOv5 for gesture detection; achieved 92% accuracy. - Integrated with OpenCV for real-time camera inference to improve accessibility solutions.	

Publication and Appearances

- **Enhancing Early Detection of Pancreatic Cancer:** A Machine Learning Approach with Explainable AI Insights – IEEE CVMI, 2024 (First Author) ([Link to paper](#))
 - > Presented research on explainable AI models for healthcare diagnostics, demonstrating applied ML pipeline development and technical communication skills.